Ziming Lin
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Education

- **University of Washington** — Seattle, US
  PhD and MS in Statistics — *Sep. 2023 – 2028 (expected)*
- **Zhejiang University** — Hangzhou, CN
  BS in Mathematics and Applied Mathematics — *Sep. 2019 – Jun. 2023*
- **University of Oxford** — Oxford, UK
  Visiting Student in Mathematics and Statistics — *Sep. 2021 – Jun. 2022*

Publications and Preprints (*: co-first author)

– Ziming Lin and Fang Han. "Generative bootstrap processes".
– Ziming Lin and Fang Han. "Bootstrap consistency for general double/debiased machine learning". **Journal of Econometrics** (Revise & Resubmit)
– Ziming Lin and Fang Han. "On the consistency of bootstrap for matching estimators". **Biometrika**
– Lars van der Laan*, Ziming Lin*, Marco Carone, and Alex Luedtke. "Stabilized inverse probability weighting via isotonic calibration". **4th Conference on Causal Learning and Reasoning** (Best Paper Award)

Presentations

- Invited Talk, Pacific Causal Inference Conference (PCIC), Beijing, July 2025

Selected Research experiences

- **Department of Statistics, University of Washington** — Seattle, US
  Research Assistant; Advised by Professor Fang Han — *Sep. 2023 - Present*
  - **Generative bootstrap processes:** Developed a general empirical-process framework for bootstrap inference based on fitted generative models and verified the theory for normalizing flows, flow matching, score-based diffusion models, and Wasserstein GANs.
  - **Bootstrap consistency of double machine learning estimators:** Established distributional consistency of the bootstrap for general sample-splitting-based procedures.
  - **Bootstrap consistency of matching estimators:** Proved bootstrap consistency for the Abadie–Imbens estimator with matches scaling in sample size. Conducted comprehensive simulation studies showing bootstrap's improved finite-sample coverage relative to existing approaches.
  - **Isotonic calibration for inverse propensity weights:** Derived the asymptotic properties of the calibrated IPW and the corresponding IC-AIPW estimators. Implemented large-scale simulations with various datasets on high-performance computing clusters.
- **Center for Data Science, Zhejiang University** — Hangzhou, CN
  Research Assistant; Advised by Professor Andre Python — *Nov. 2021 - Apr. 2023*
  - Applied interpretable ML approaches to support drug discovery and malnutrition detection in cancer patients, gaining hands-on experience with ensemble methods and efficient hyperparameter tuning.
- **Department of Statistics, University of Oxford** — Oxford, UK
  Research Assistant; Advised by Professor Geoff Nicholls — *Apr. 2022 - Aug. 2022*
  - Explored variational inference and flow-based generative models for discrete distributions, and developed theoretical grounding in generative models and nonparametric methods.

Skills

Programming & Tools: Python, R, C, HPC (Slurm, Bash) | **Languages:** English, Mandarin
Selected Coursework: Statistical Machine Learning, Deep Learning, Causal Inference, Stochastic Processes, Natural Language Processing, Convex Optimization, Design and Analysis of Experiments, Simulation Methods, Bayesian Methods, Advanced Regression Methods, Empirical Process Theory, High-Dimensional Statistics